

Problem 10.39

Find δ_t if

$$A(t) = K2^t 3^{t^2} 5^{2^t}.$$

Solution

The force of interest is

$$\delta_t = \frac{A'(t)}{A(t)} = \frac{d}{dt} \ln A(t).$$

Take the natural logarithm of the accumulation function:

$$\ln A(t) = \ln \left(K 2^t 3^{t^2} 5^{2^t} \right).$$

Using the logarithm rules,

$$\ln A(t) = \ln K + t \ln 2 + t^2 \ln 3 + 2^t \ln 5.$$

Differentiate with respect to t :

$$\delta_t = 0 + \ln 2 + 2t \ln 3 + \frac{d}{dt}(2^t) \ln 5.$$

Since

$$\frac{d}{dt}(2^t) = 2^t \ln 2,$$

we obtain

$$\boxed{\delta_t = \ln 2 + 2t \ln 3 + 2^t \ln 2 \ln 5}.$$